

SiGe Qubit Biasing with a Cryogenic CMOS DAC at mK Temperature

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Abstract— For running advanced algorithms on a universal quantum computer, millions of qubits are required. To make use of quantum effects, state-of-the-art solid-state qubit devices have to be cooled to mK temperatures, which limits the systems' scalability with room temperature (RT) electronics. We present the direct co-integration of a scalable, fully integrated, eight channel Bias-DAC designed in a 65-nm bulk CMOS technology and a Si/SiGe spin qubit device at the mixing chamber (MC) of a dilution refrigerator operating below 45 mK MC temperature. As a full proof of principle, the bias of a single electron transistor used as a sensing dot, as well as a single and double quantum dot bias of the qubit device is reported. The slow drift of the DAC S&H output circuit of 0.96 $\mu\text{V/s}$ leads to a calculated prospective power consumption of 64.5 pW/ch for DC qubit bias voltages generated at low temperature.

Index Terms—CMOS, cryogenic temperature, quantum computing, spin qubit, cryogenic CMOS, DAC

I. INTRODUCTION

Quantum Computing could circumvent the current speed limits in classical state-of-the-art computing. Certain classical computing problems, such as the prime factorization of large numbers, could become solvable due to the exponential reduction of classical computing time through quantum computing [1], [2]. Instead of classical bits, quantum computers use quantum bits (qubits) which allow for a coherent superposition of the state 0 and 1 during computation. In general, every two-state quantum system which fulfills the DiVincenzo's criteria could be used to implement a physical qubit [3]. The most advanced approaches are superconducting circuits, trapped ions, neutral atoms and electrons in semiconductor materials [4]–[8]. To observe quantum effects like superposition or entanglement, the qubit device has to be cooled to mK temperatures with the help of dilution refrigerators (DR) to suppress thermal excitation. Additionally, qubits require dedicated electronics for the generation of control signals and read-out, which are currently located outside the cryostat for state-of-the-art quantum processors. Increasing the qubit number requires a large number of wires going into the DR connected to these RT electronics. Brute force scaling approaches are promising for NISQ applications with up to

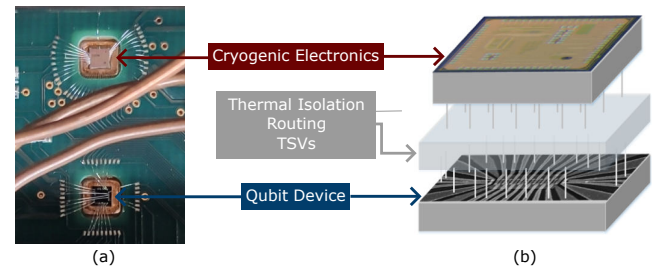


Fig. 1. Placement of cryogenic electronics (co)-integrated with a qubit device. (a) Approach of this work: Top view of IC and Qubit device placed next to each other on MC (b) Possible future heterogeneous 3D integration.

about 500-1000 physical qubits, but are limited towards further upscaling due to wiring limitation inside state-of-the-art DRs [9]. Universal quantum computers require qubits in the scale of millions to implement systems, including the necessary error correction [10]. Furthermore, every cable increases the heat load. Thus, extremely careful engineering already at the 100 qubit level is required [11]. Cryogenic CMOS electronics can solve this bottleneck. Instead of a linear increase of the wires per qubit the CMOS integrated circuit (IC) wiring for supplies and communication do not increase with the same factor as the number of qubits. For a highly scalable system, spin qubits in semiconductor quantum dots are a natural choice due to their relatively robust coherence, small spin dimension, and good compatibility with CMOS-generated control signals. Due to a purely capacitive load of spin qubits in semiconductor, the opportunity for heterogeneous 3D integration of quantum and classical chips in the future is opening up (See Fig. 1).

The cooling power of state-of-the-art DRs at 100 mK are stated in the range of hundreds of μW and dictates a strict power dissipation limit for the cryogenic ICs. In order to avoid the tight power dissipation limit on the dilution mixing chamber (MC) stage, some approaches place the qubit control electronics at the intermediate 4K stage [12], [13]. To circumvent the resulting wiring bottleneck from 4K to the MC, including the thermal anchoring challenges of the wiring, a fully

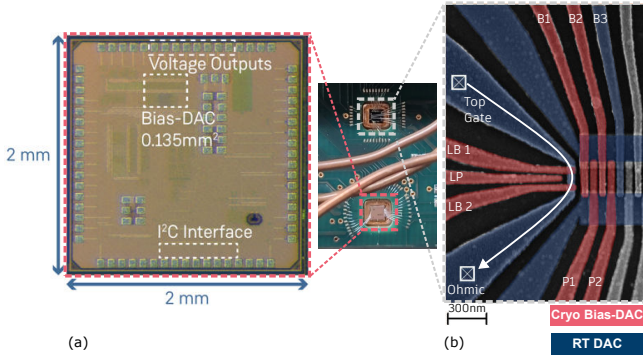


Fig. 2. (a) Die picture of the IC (b) Top view of qubit device. The gates biased by the Bias-DAC are marked red. The implanted ohmic contacts are indicated by the two crosses.

co-integrated solution at mK next to the qubit is considered here. Spin qubits in SiGe require two types of control signals, DC-tuning-voltages to ensure the correct operating point and RF control signals. Control signals might be shared between multiple qubits in future architectures, or could potentially be generated with fast DACs. As a first essential step, we focus on the DC-signal generation. Two competing approaches have been demonstrated to work for this task. Generation of voltages at room temperature and demultiplexing to sample-and-hold (S&H) capacitors at cryogenic temperatures relies on the diminishing leakage currents in the mK regime [14]. Still, necessary refresh rates might place a limit on operation speed and gate number. Alternatively, voltages can be generated by an integrated DAC autonomously at mK temperatures, only requiring communication to room temperature for voltage changes [15], [16]. The increased complexity of the digital circuit however increases the dissipated power on the MC.

Here, we report the co-integration of a Bias-DAC next to a Si/SiGe qubit device at the mK stage of a DR with direct voltage generation at the qubit layer. The DAC has a power consumption of $20 \mu\text{W}$ and with $0.96 \mu\text{V/s}$ a small voltage drift over time. We present a fully functional single electron transistor (SET) bias used as a sensing dot and charge sensing of a single and double dot bias with the DAC.

II. SETUP

A. Qubit device

This work focuses on electrons in quantum dots formed in a two dimensional electron gas (2DEG) in a Si/SiGe heterostructure [17]. The qubit information is stored in the spin of such an electron captured in the quantum dot. Forming quantum dots requires DC bias applied to metal gates on top of the heterostructure to accumulate and deplete electrons locally in the quantum well, i.e. approx. 32 nm below the surface of the structure. A possible architecture is discussed in great detail in [18], [19]. An oxide layer between the gates and the substrate reduces leakage through the device, hence the load is mainly capacitive. Additionally, ohmic contacts overlapping with the top gate are implanted to be able to apply a voltage

TABLE I
TEMPERATURE DISTRIBUTION

Sensors	Bias-DAC off	Bias-DAC on
Rox MC	13 mK	44 mK
Rox Sample Holder	$< 50 \text{ mK}^a$	180 mK
Electron Temp Quantum Dot	-	$\sim 400 \text{ mK}$

^aRuthenium oxide sensor calibrated from $50 \text{ mK} - 40 \text{ K}$.

bias to the accumulated electrons in the 2DEG and measure the transport current through the SET, used as a capacitive coupled sensing dot (see Fig. 2b).

B. CMOS Bias-DAC

The Bias-DAC is integrated on a 2 mm^2 CMOS chip (see Fig. 2a). It is based on the charge redistribution topology and designed in 65 nm bulk CMOS technology. For a more detailed description see [15]. The DAC uses a 10 bit split capacity array and additionally eight switchable reference voltages for coarse tuning, increasing the resolution to overall 13 bit with an output voltage range from 0 V to 1 V . The resulting step size is about $122 \mu\text{V}$. The output voltage is demultiplexed to eight independent and simultaneously working S&H structures with two 1 pF capacitors. These are bedded into two-stage RC filters with a cutoff frequency of 6.4 MHz and 1.06 MHz , below the clock frequency range of the DAC to attenuate a possible feed through. Seven output channels are routed via the PCB to the qubit device and are wire-bonded to the gate electrodes. The eighth one is routed to RT in order to characterize the DAC performance. Due to possible leakage of the S&H capacitors, a periodic refresh of the output channels is required. In previous work, the DAC was tuned to best working performance at a refresh rate of 3894.1 Hz and a lowest stable clock frequency of 10 MHz to reduce the dissipated power [16].

C. Integration

The first integration with a GaAs qubit device showed promising results regarding the DAC performance with the qubit but left potential in thermal decoupling of qubit and electronics [16]. In this work, a PCB with two rectangular cut-outs is used and the Bias-DAC and the qubit device are placed in direct metal contact with the mixing chamber plate. The electrical connections are done with $25 \mu\text{m}$ aluminum bond wires to bond pads on the PCB. Low impedance lines are used for the 2.5 V and 1.2 V DAC supply, as well as digital communication signals. The DAC is connected to the left SET gates (LB1, LP, LB2) and the first two barrier and plunger gates (B1, B2, P1, P2) of the shuttling architecture (see Fig. 2). The accumulation with the left top gate is done with an RT voltage source. The current trough the 2DEG is sensed at RT with a TIA (Basel Precision Instruments SP983c-IF) and measured with a PCI-e data acquisition card (Alazar Tech ATS9440).

III. MEASUREMENT RESULTS

The measured total power consumption of the IC is about $40 \mu\text{W}$, which is due to additional testing circuits integrate on

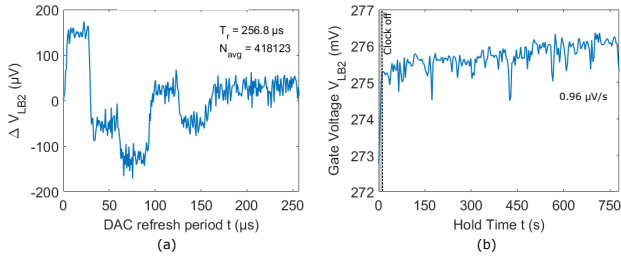


Fig. 3. Bias-DAC Measurements via current measurement through the SET. (a) Signal average over S&H refresh period re-mapped to voltage changes at gate LB2. Eight times faster periodically switching related to the DAC outputs are indicated. (b) Leakage measurement after clock turn-off reconstructed to voltage drift of gate LB2

the IC that cannot be completely switched off, but are not required for the Bias-DAC. The power consumption of the Bias-DAC itself of about $20 \mu\text{W}$ is in agreement with earlier results [15], [16]. To clarify that the DAC is not limiting the qubit operation due to thermal heat dissipation, the temperature at the MC and the backside of the copper sample holder are measured via ruthenium oxide sensors (Rox). The DAC operates at 44 mK MC temperature and 180 mK sample holder temperature. The electron temperature in the quantum dot of the qubit device can be calculated from a measurement of the charge transition of the dot. With reference to results presented later in section III-A, we can roughly estimate an electron temperature of 400 mK (see Tab. I). We can assume, that this temperature is in the range in which qubit operations become possible [20], [21].

After the accumulation of electrons in the 2DEG below the top gate, the functionality of the SET gates biased with the DAC is tested via transconductance sweeps. For qubit operations, leakage currents are a limiting factor, but typical RT amplifiers draw larger leakage currents than the leakage to be investigated. Hence, it must be measured in the operating environment. The qubit device itself can be used as a measurement characterization device for cryogenic ICs. To achieve a conductance measurement sensitive to the voltage changes of the DAC outputs, one barrier gate of the SET is set to the steepest point of the edge of the gate transconductance i.e. gate LB2. The periodic refresh over time is the region of interest for the leakage measurement. Therefore, a repetitive average is done at $f_r = 3894.1 \text{ Hz}$ (see Fig. 3a). We observe voltage jumps with an eight times faster frequency than the refresh rate which can be identified as the LB2 refresh and cross-talk of the consecutive refreshes of the overall eight S&H structures. A maximum voltage ripple of $\Delta V_{pp} = 275 \mu\text{V}$ is measured. Influences of the other gates to the measured conductance are minimized by operating at a point with minimal transconductance. During the hold time of each particular output, no leakage is observed.

To measure the potential voltage drift due to leakage without a refresh, the clock is turned off at the beginning of the hold time measurement. During an overall measurement time of 13 min, we measure a voltage drift of $749 \mu\text{V}$ resulting in

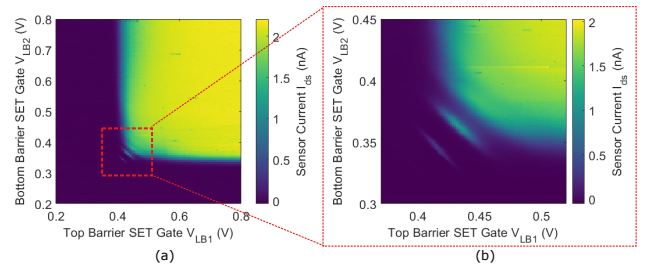


Fig. 4. Coulomb Oscillations at the SET of sensing quantum dot. (a) Wide scan to identify the coulomb blockade region. The steep pinch-off of the conductance channel verifies capable gate control. (b) Scan of the region of interest is showing clear coulomb oscillations.

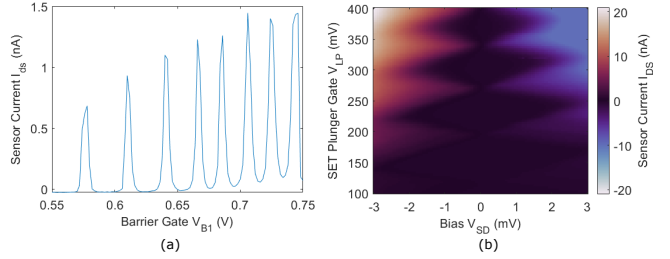


Fig. 5. Gate B1 sweep (here used as the SET plunger) shows sharp coulomb peaks, as well as zero current regions between the three lowest peaks. (b) Sweeping the SET plunger gate while increasing the applied bias result in characteristic coulomb diamonds. A larger bias increases the amount of discrete energy levels in the dot, which leads to a broadening of the coulomb peaks.

$0.96 \mu\text{V/s}$ (see Fig. 3b). With a voltage stability requirement at a qubit gate of $10 \mu\text{V}_{pp}$ [19], the gate refresh rate of the DAC can be extrapolated to a possible frequency of 0.1 Hz. With this frequency and with respect to the measured voltage drift of $0.96 \mu\text{V/s}$, we can theoretically bias up to $\sim 310,000$ gates with a single Bias-DAC. The measured DAC power consumption concludes to an extrapolated power consumption of 64.5 pW/ch .

A. Qubit Device Measurements

As a proof-of-principle, typical SET measurements and charge sensing of a single and a double quantum dot biased with the Bias-DAC are presented. Both barrier gates (LB1, LB2) show steep pinch offs curves. We observe characteristic coulomb oscillations in the SET by sweeping the two SET barrier gates and with that the electrochemical potential of the SET step-wise to the region where only tunneling from the reservoir to the discrete levels of the quantum dot is possible (see Fig. 4) [22]. Depending on the applied bias window only individual electrons are contributing to the current flow and due to the discreteness conductance peaks are visible. These coulomb peaks broaden with the applied voltage bias to characteristic coulomb diamonds (see Fig. 5) [23]. By setting the sensing dot in the SET to a sensitive region, the charge configuration from the adjoined formed quantum dot underneath P1 can be detected (see. Fig. 6). Jumps in the

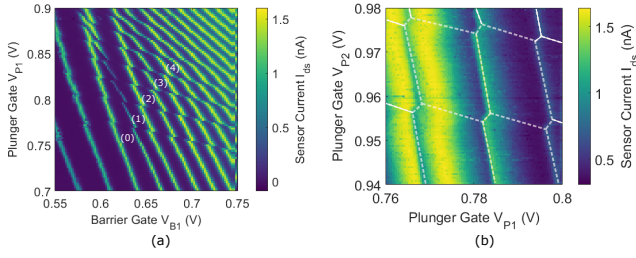


Fig. 6. Charge sensing of a single and double quantum dot. (a) Single dot formed underneath P1. The numbers indicate an additional electron hopping into the dot from the reservoir. (b) Double dot formed underneath P1 and P2. The inter-dot transition is clearly visible, likewise the characteristic honeycomb pattern.

sensor peaks indicate an additional electron hopping into the quantum dot initiated by lowering the barrier gate towards the SET used as an electron reservoir [24]. Additionally, a double quantum dot underneath P1 and P2 with a visible inter-dot transition and the typical honeycomb pattern is observed. Controlling the exact number of electrons occupying the dot is essential for further spin manipulations. Combined with the considerations of a timed and controlled turn-off of the DAC during qubit operations, feasible due to the observed low leakage, the presented Bias-DAC achieves a stable DC qubit bias.

IV. CONCLUSION

The co-integration of a bulk CMOS eight channel Bias-DAC with a Si/SiGe spin qubit device at 44 mK MC temperature has been demonstrated. Sharp coulomb peaks with intermediate zero-current regions in the coulomb blockade regime, the measurement of the electron temperature and the measured temperature at the sample holder of 180 mK indicate, that the influence of the thermal heat dissipation of the IC, while operating next to the qubit device can be narrowed down to a range in which qubit operations become possible. With respect to the low voltage drift of $0.96 \mu\text{V/s}$ at the DAC output over time, we can extrapolate the power consumption per channel towards 64.5 pW/ch . Despite the low power budget at the qubit layer, the presented DC qubit bias with sufficient charge sensing of electrons in the quantum dot determine the feasibility of cryogenic low power electronics at the lowest temperature stage of an DR.

V. ACKNOWLEDGEMENTS

The qubit device fabrication has been done at HNF - Helmholtz Nano Facility, Research Center Jülich GmbH [25]

VI. AUTHOR CONTRIBUTIONS

L.S. and R.O. contributed equally to this work. P.V. designed the DAC. R.X., I.S., J.-S. T., S.T. and L.R.S. worked on the fabrication of the qubit device. L.S. wrote the paper with input from all other authors. H.B. and S.v.W. guided the work.

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